MIL-E-14776A(PA)
23 November 1973
SUPERSEDING
MIL-E-14776(MU)
12 June 1968

MILITARY SPECIFICATION

EXTENSION, FIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of fin extension for use with 60MM Mortar Cartridges designated as Extension, Fin.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons General Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1169	-Packaging, Packing and Marking, for Shipment of Inert Ammunition Components
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9215574	-Extension, Fin

FSC: 1310

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Fin Extension.-The fin extension shall comply with all requirements specified on Drawing (dwg.) 9215574 and with all requirements specified in applicable specifications.

3.3 Workmanship.-The fin extension shall be fabricated and finished in a thorough workmanlike manner and shall be free from dirt, grease, chips, oil, corrosion or other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

3.4 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.

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e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized, by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed.-Parts will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any part fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part in the sample to comply with any or all of the stated requirements.

4.3 Inspection Provisions

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4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

4.3.2.1 Extension fin (see dwg. 9215574)

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
101.	Concentricity of internal thread to external thread including perpendicularity to face.....	Gage (CD)

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102. Depth of internal threads, min.....Gage (CD)
103. Diameter of external thread undercut..Gage (CD)
104. Length of external thread to shoulder, max.....Gage (CD)
105. Pitch diameter of external thread, min.....Gage (CD)
106. Major diameter of external thread, min.....Gage (CD)
107. Pitch diameter of internal thread, max.....Gage (CD)
108. Minor diameter of internal thread, max.....Gage (CD)
109. External threads not full for entire circumference of perfect thread length.....Visual
110. Internal threads not full for entire circumference of perfect thread length.....Visual
111. Length of outside diameter.....Gage (CD)

Minor:
201. Concentricity of outside diameter to external thread including perpendicularity to shoulder.....Gage (CD)
202. Outside diameter.....Gage (CD)
203. Depth of internal thread cavity, max..Gage (CD)
204. Fillet radius of external thread undercut, max.....Gage (CD)
205. Length of external thread undercut, max.....Gage (CD)
206. Finish improper.....Visual-Tactile
207. Nib length.....Gage (CD)
208. Evidence of poor workmanship (see 3.3)Visual

4.3.3 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 (see 6.2).

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.-Packing and marking shall be in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

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a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, that established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design " for guidance. The specification number, paragraph number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing and revision letter to which the specific design applies.

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6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Submission of initial production samples.-Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SARPA-QA.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1310-A048

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: *The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document.* DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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